

Grundlagen Digitaler Medien

S3D Video - Grundlagen

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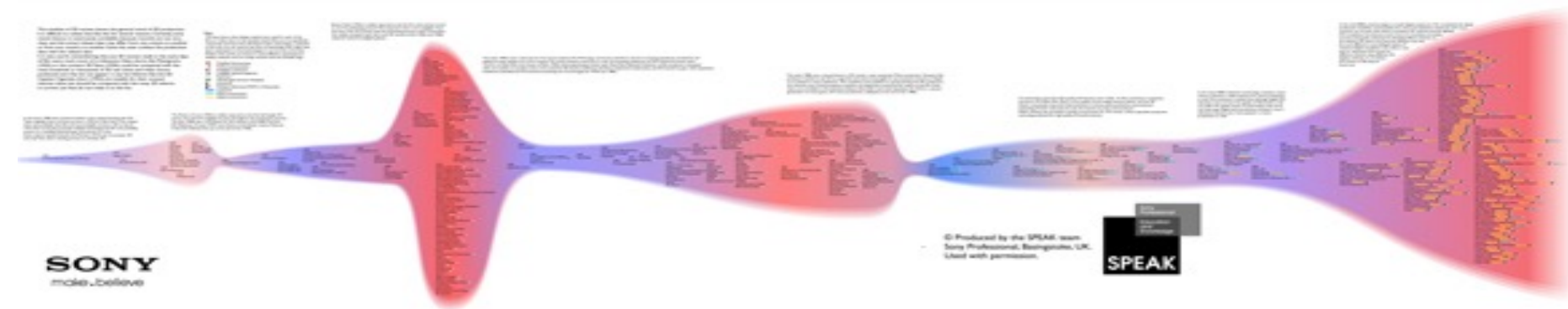
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Lernziele

- Verständnis für die Grundlagen der Stereoskopie
- Verständnis für die Darstellung von Stereo-3D Inhalten und die verschiedenen Display-Technologien

The 3D Ages

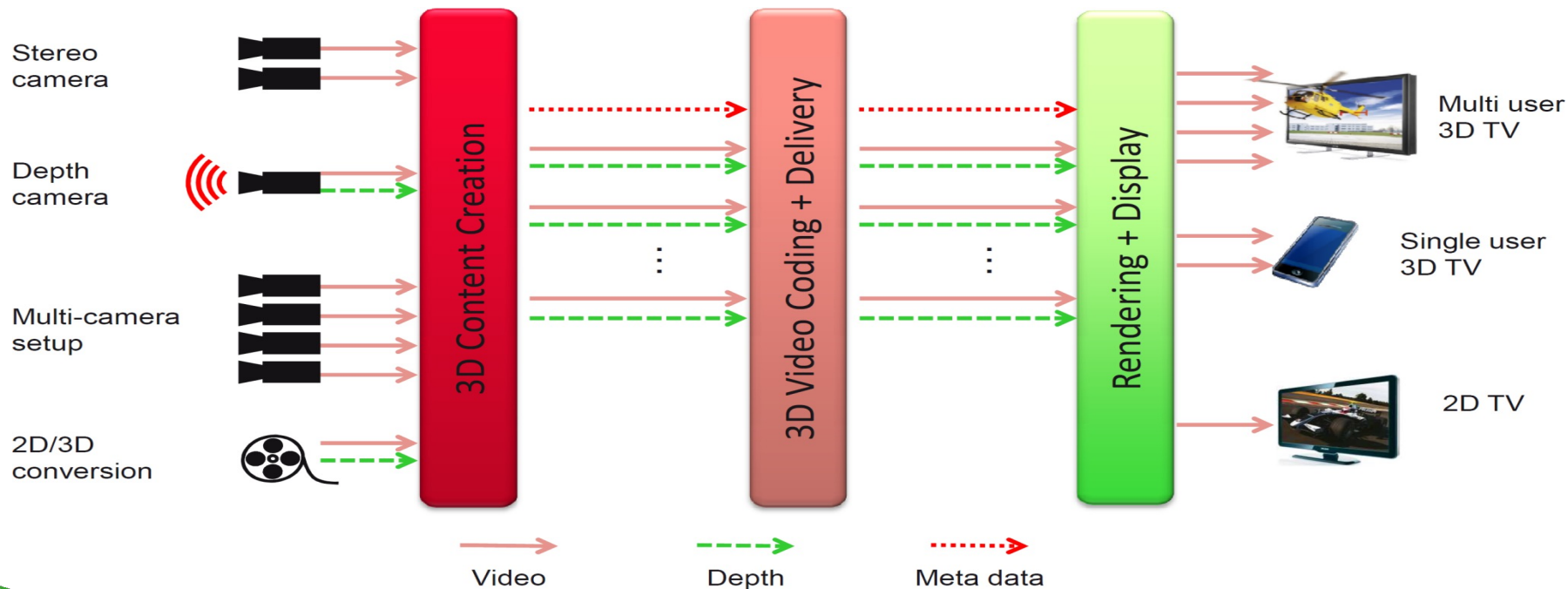


1950's 1970's & 1980's

2001 until today



3D Prozesskette



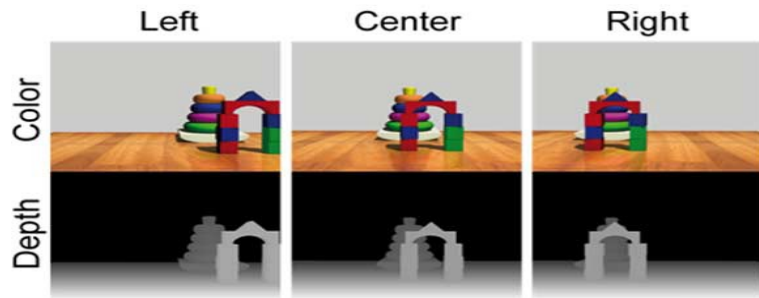
3D Content Creation

- Computer generated imagery (CGI)
- **Stereoscopic camera**
- **2D/3D conversion**
- Image+depth sensors
- Multi camera setups

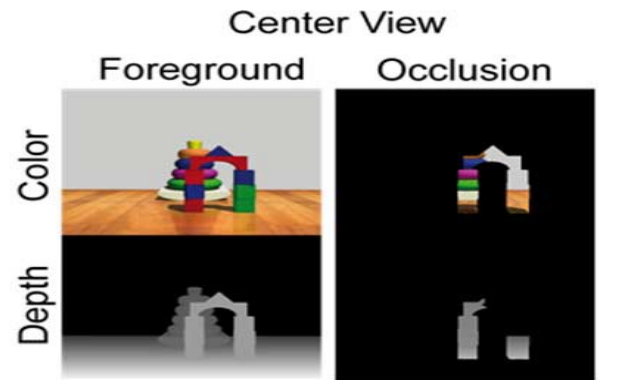


3D Content Coding & Transmission

- Video+depth (VD)
- Multiview (MV)
- Multiview video+depth (MVD)
- Layered video+depth (LVD)



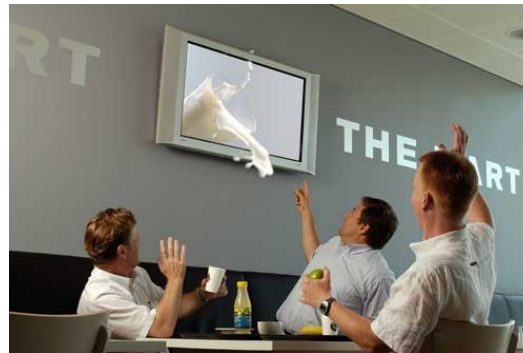
Multi Video plus Depth



Simplified Layered Depth Video

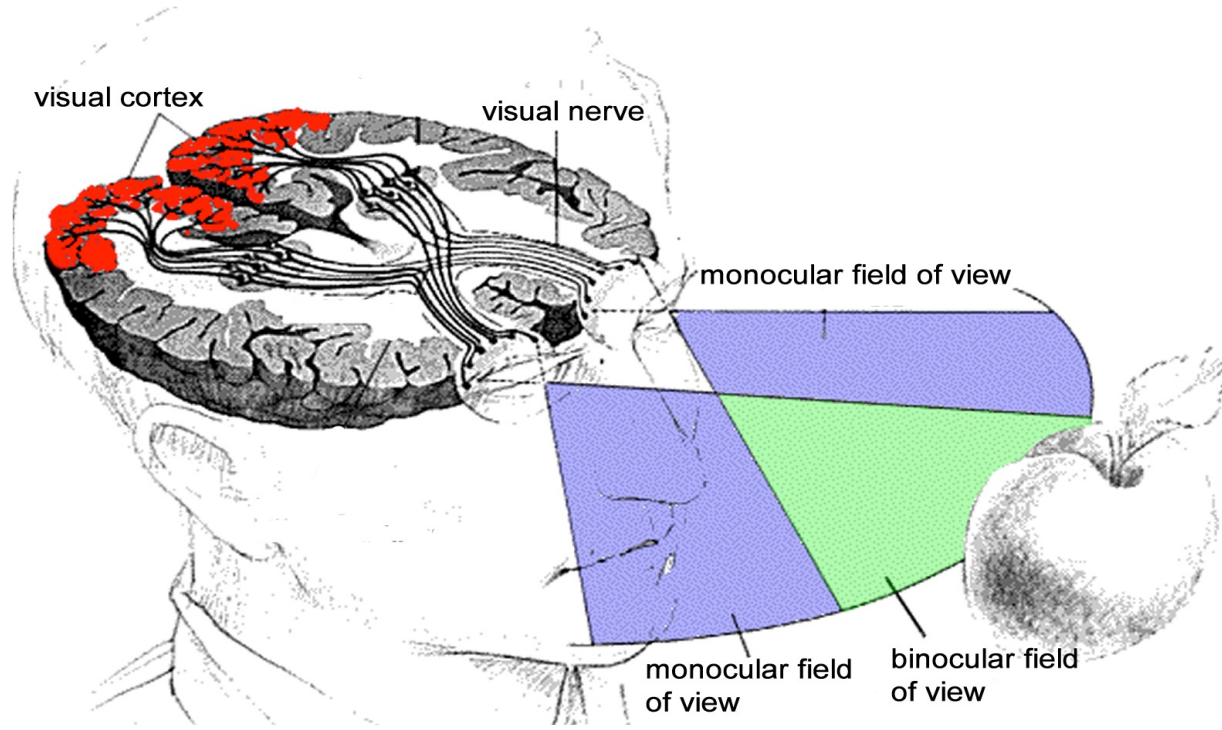
3D Content Display

- **Stereoscopic: anaglyph, polarized, shutter, wavelength multiplex**
- **Autostereoscopic: lenticular lens, parallax barrier**
- Integral imaging
- Holographic
- Volumetric



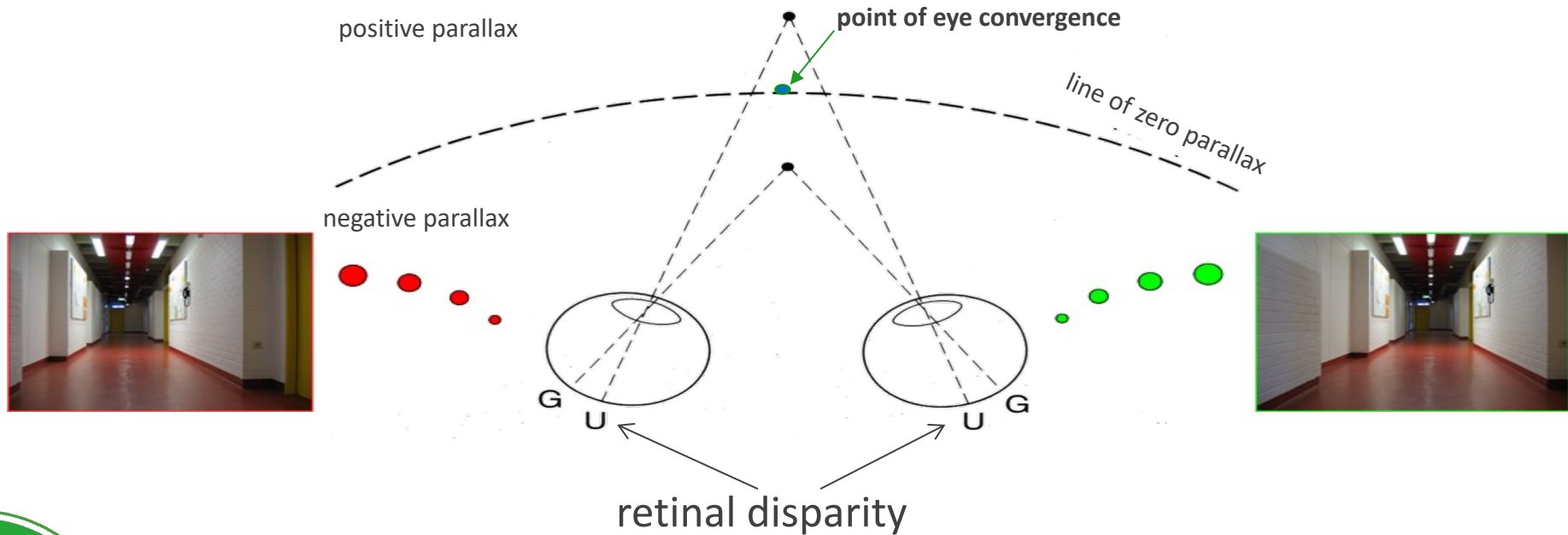
Grundlagen der Stereoskopie

Rückblick: Human Visual System (HVS)

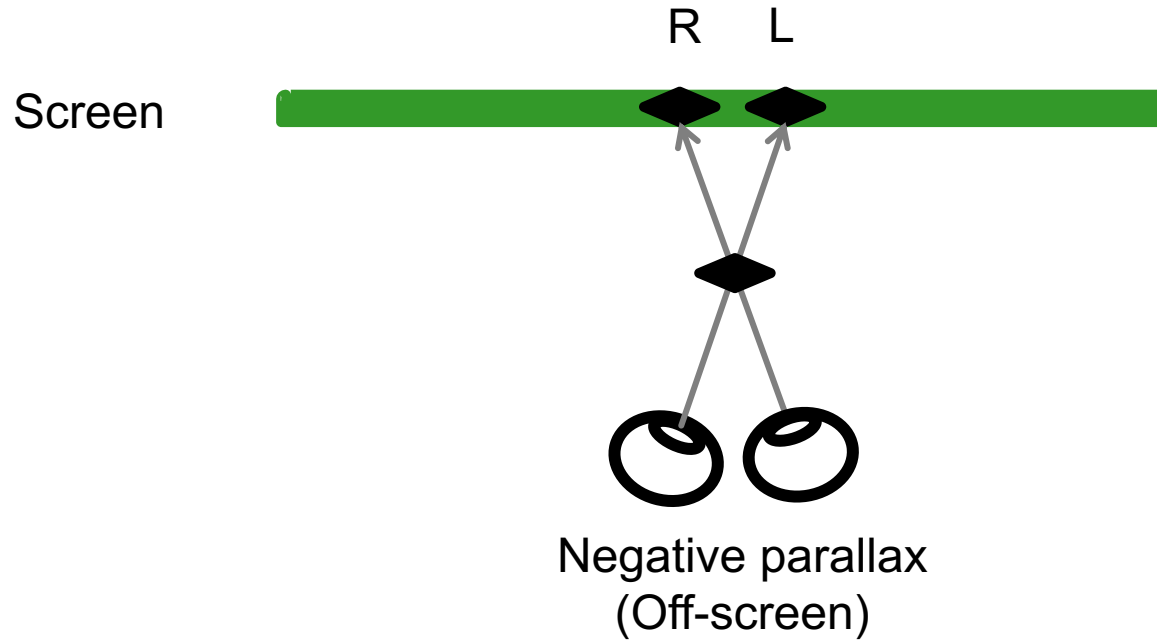


Rückblick: Stereoskopie

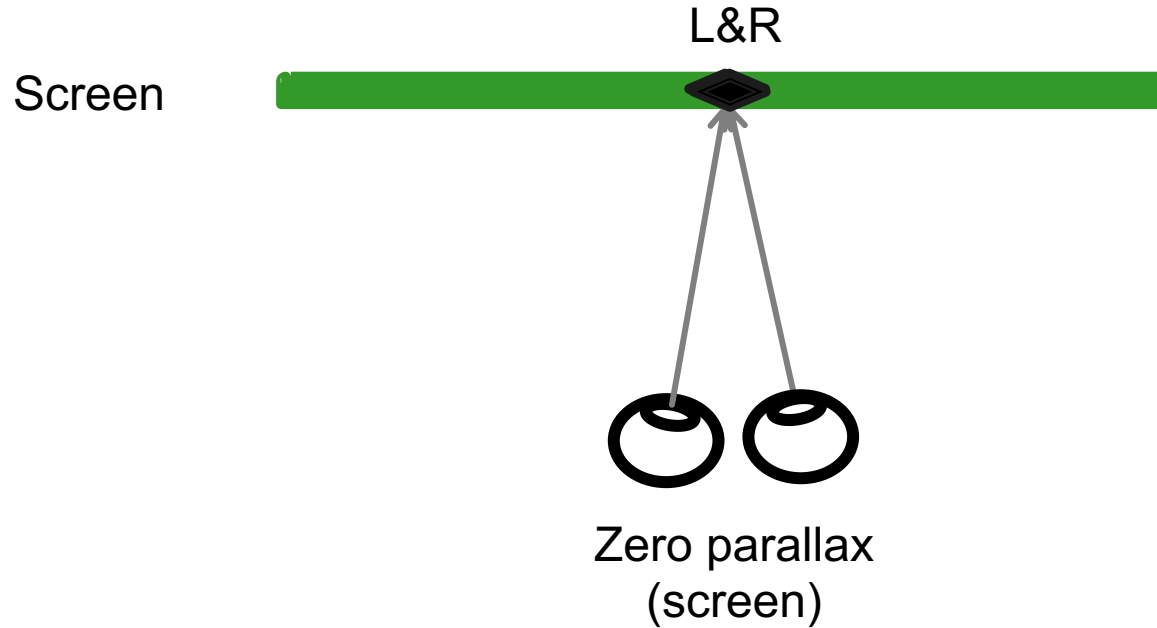
Vergence and stereopsis:



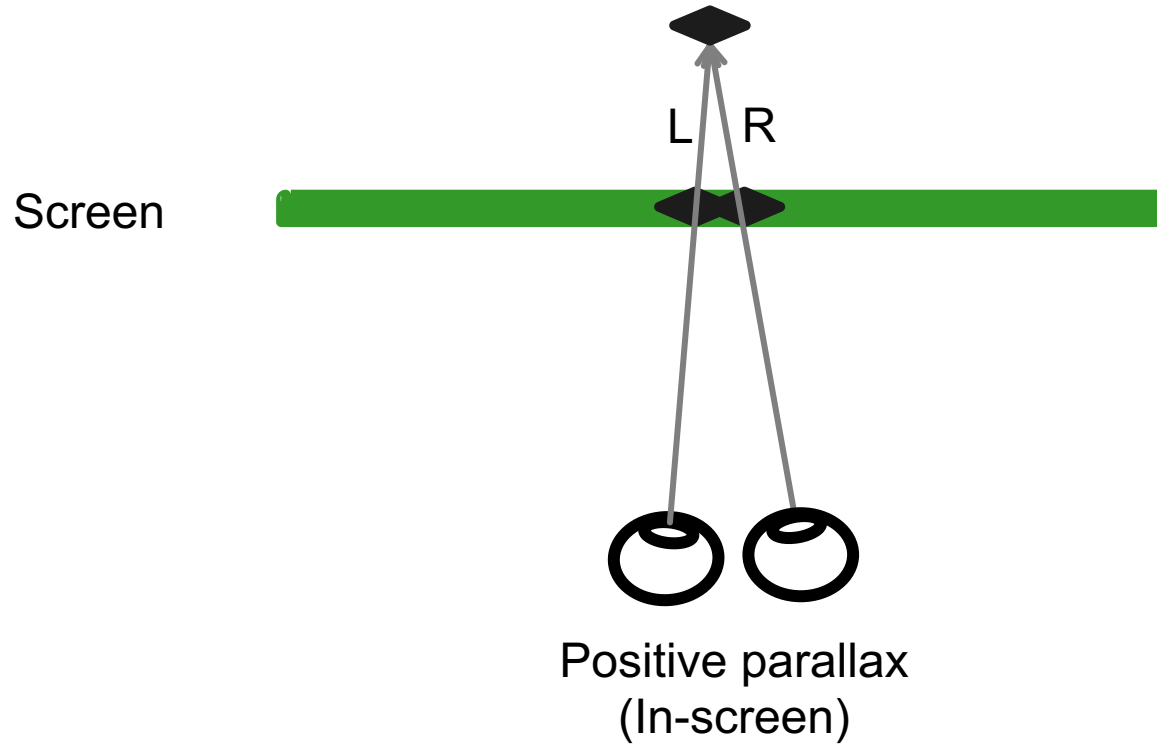
Stereoscopic 3D – off screen



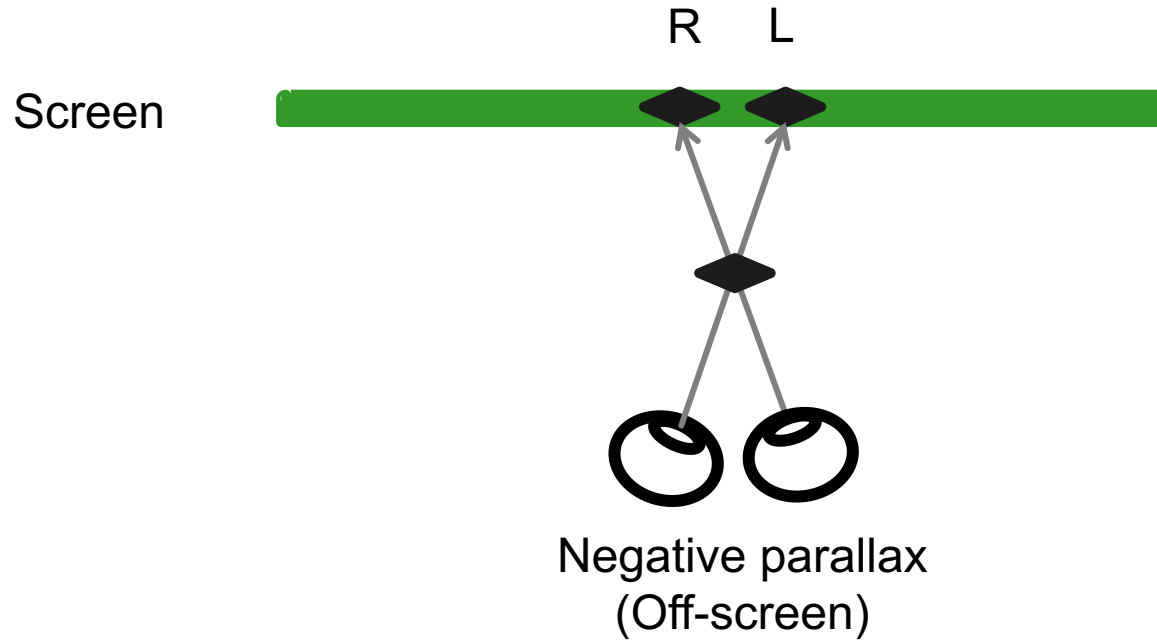
Stereoscopic 3D – on screen



Stereoscopic 3D – in screen



Stereoscopic 3D – off screen





Frame Violation

Looking through a real window with both eyes open

not visible with right eye



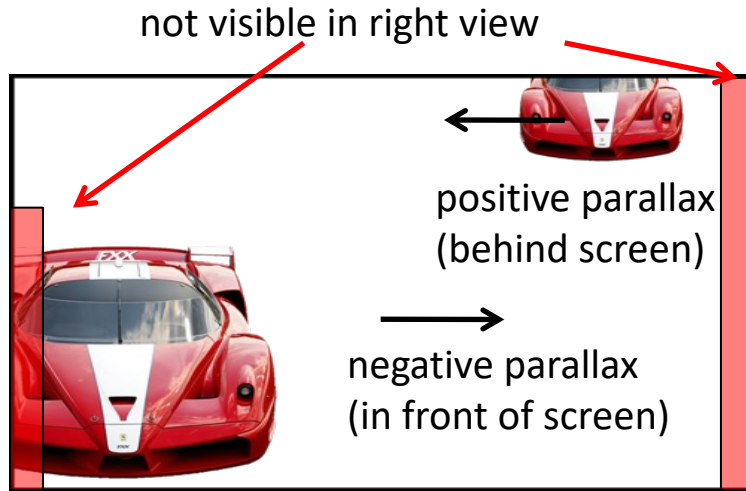
left eye view

not visible with left eye

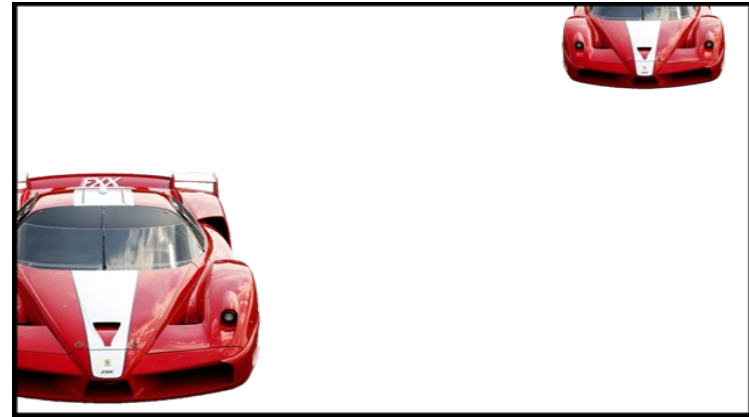


right eye view

Frame Violation



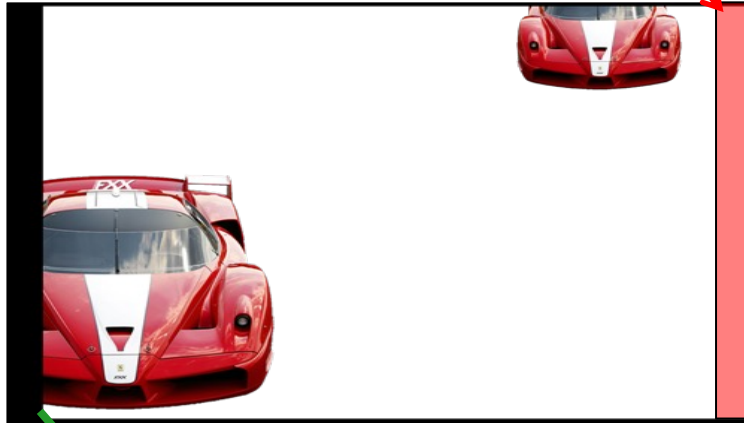
left image



right image

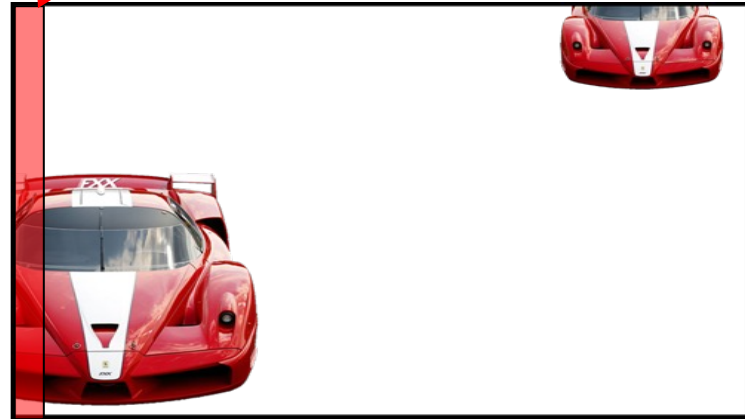
Floating Window

not visible with right eye
left image

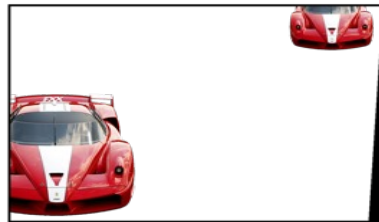
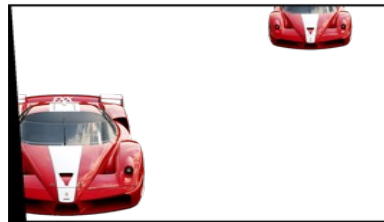
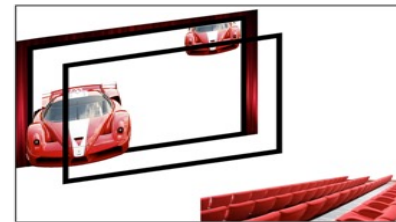
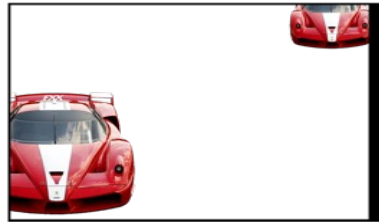
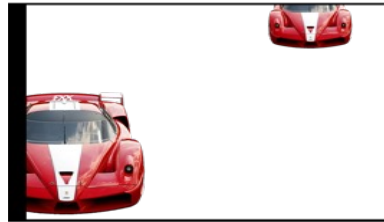
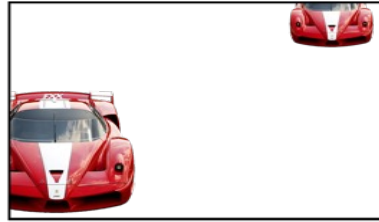
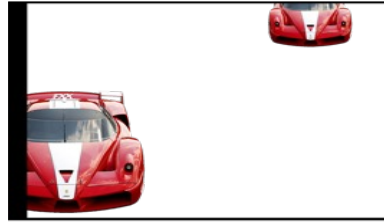


floating window

not visible with left eye
right image



Floating Window

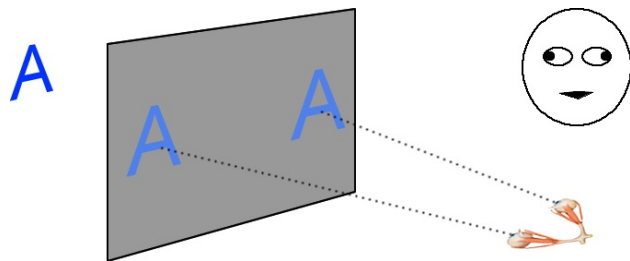


Left image

Right image

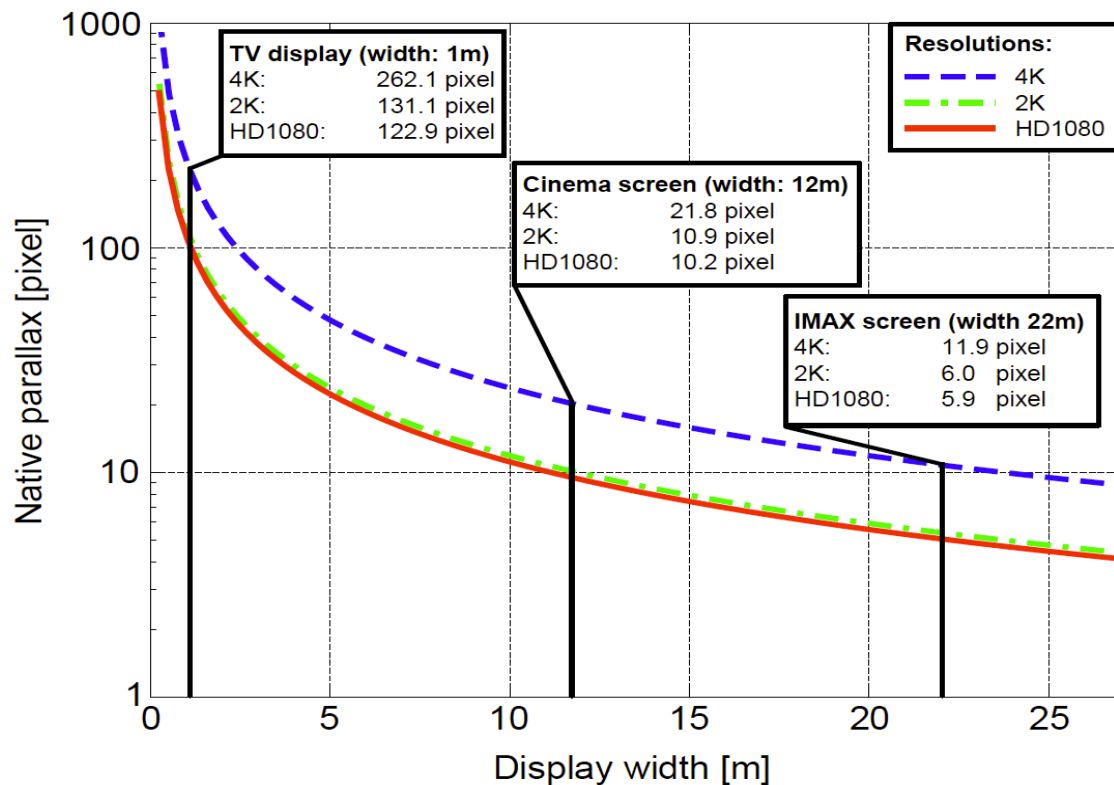
Appearance on
the screen

Native Pixel Parallax and Depth Budget

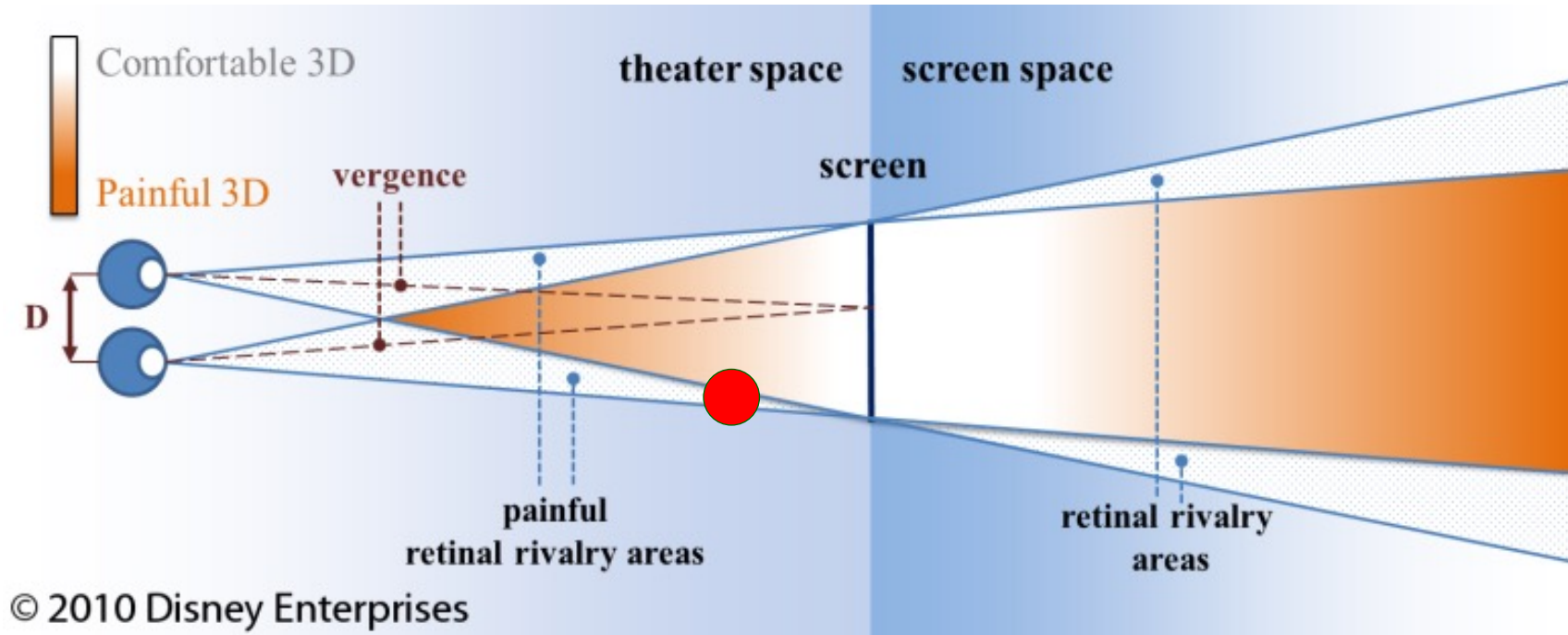


Depth budget:

	Foreground (- parallax)	Background (+ parallax)
IMAX	1.5 – 3 %	< 0.25 %
Cinema	1 – 2.5 %	< 1 %
TV	0.5 – 1.5 %	1.5 – 2.5 %

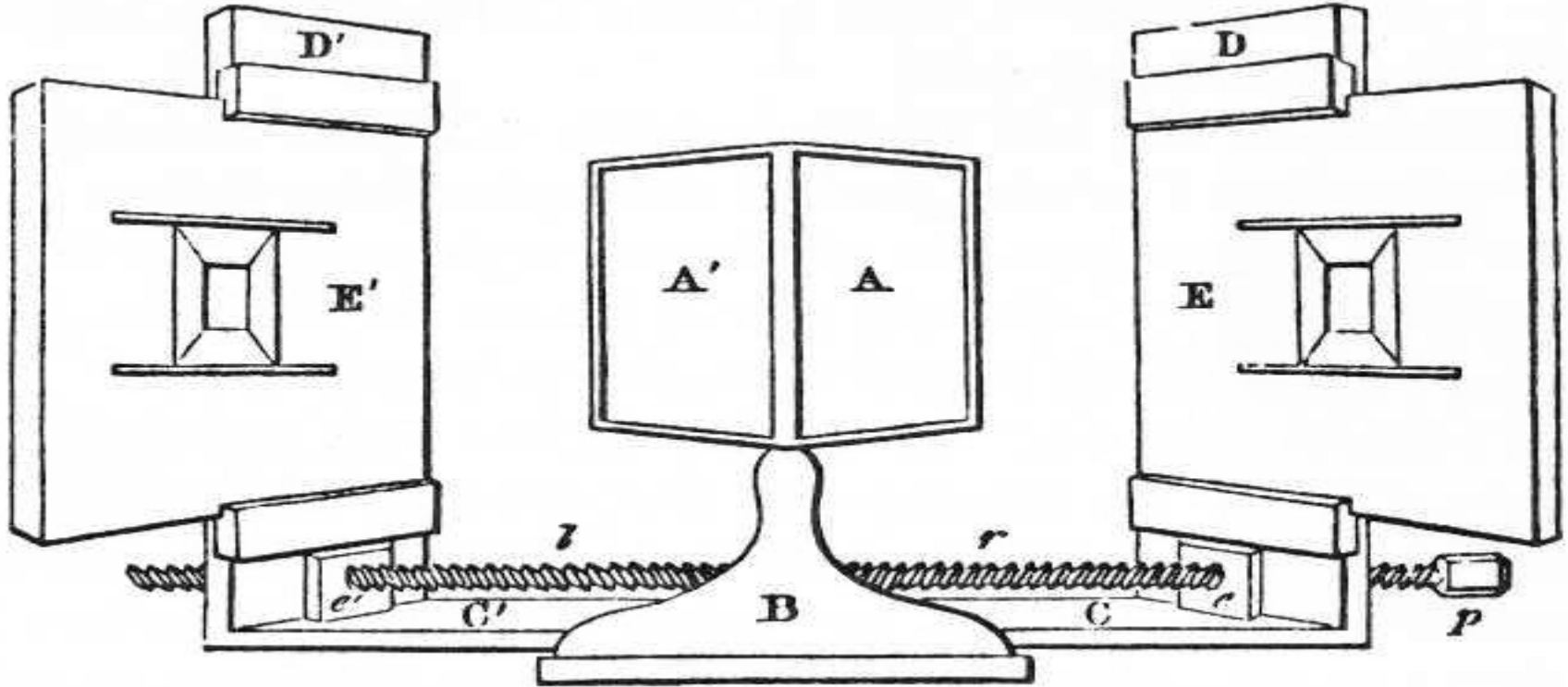


3D Comfort Zone

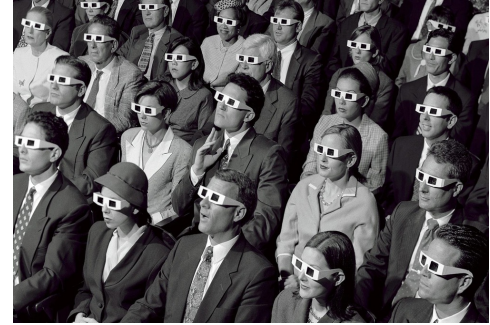
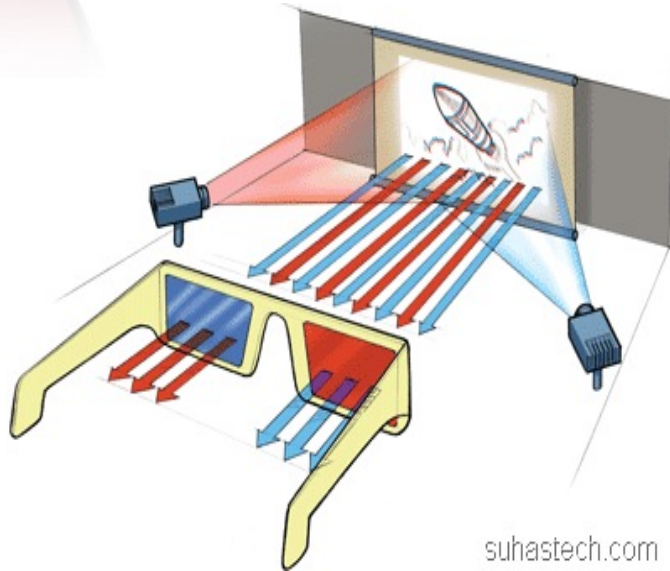


S3D Display

Wheatstone Stereoscope (1838)



Anaglyph 3D Technology (1950s)

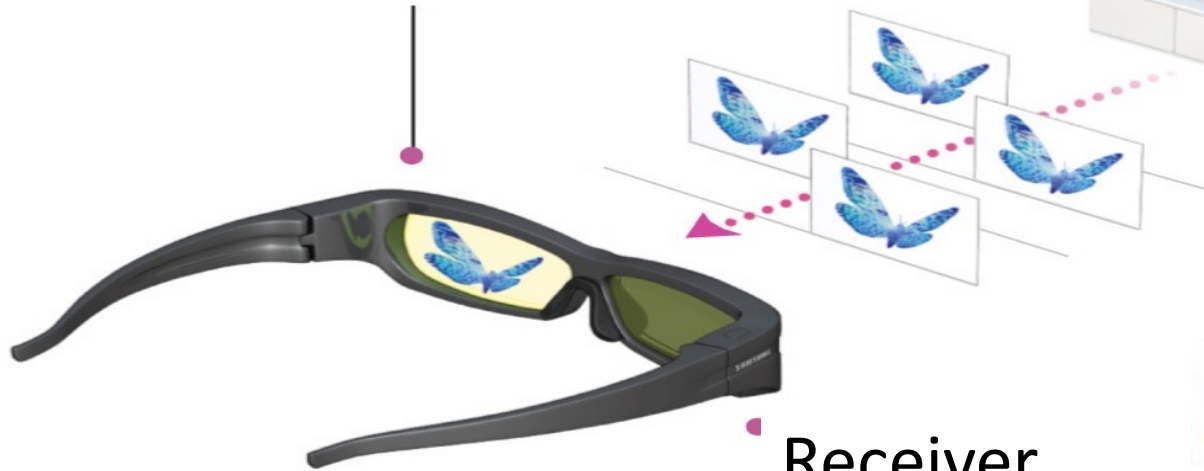


3D Shutter Technique

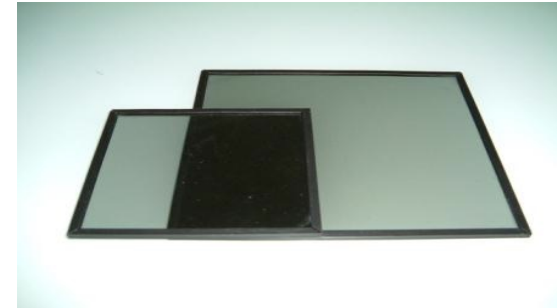
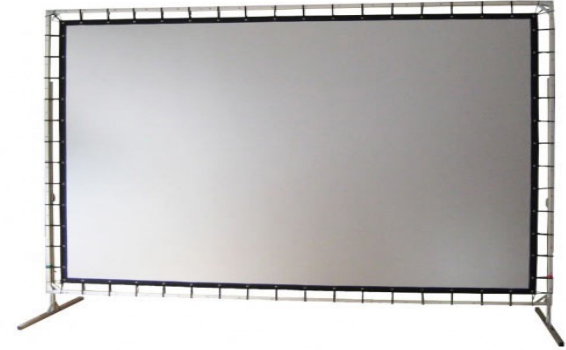
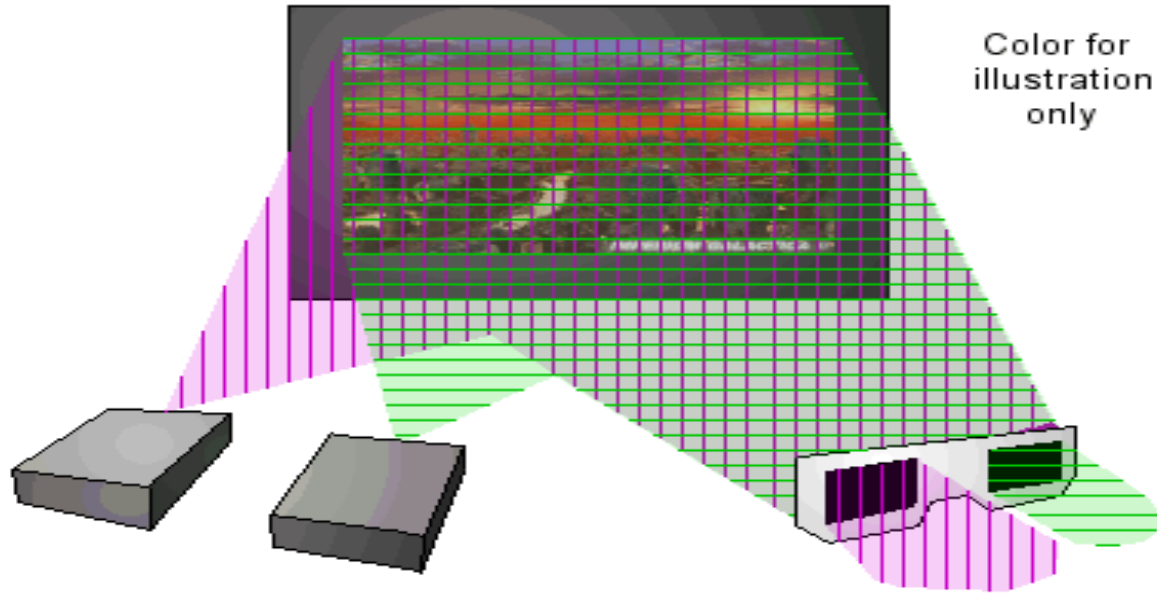
L R L R

Time-sequential left/right

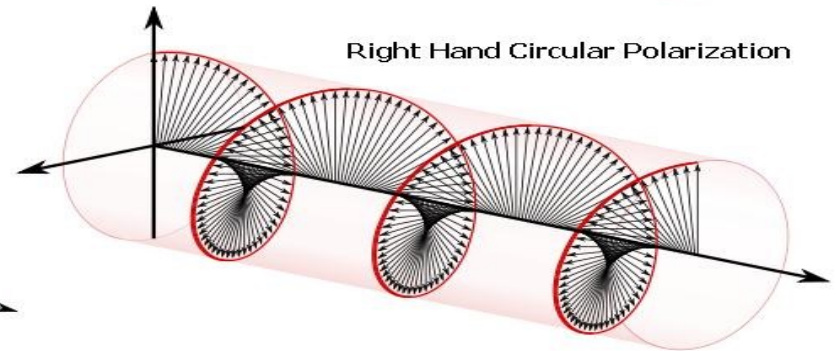
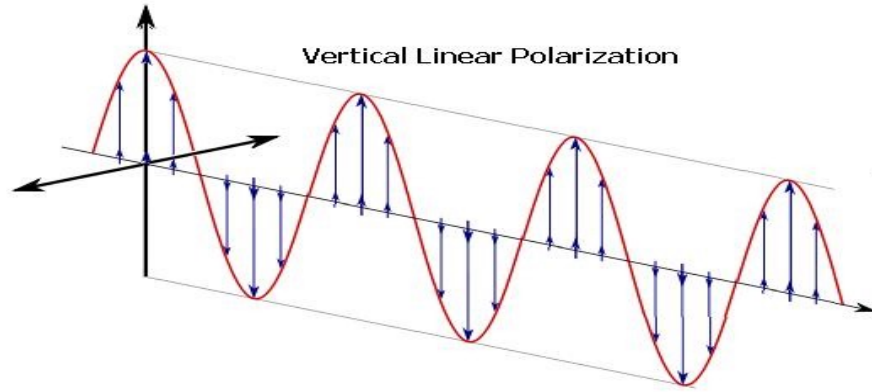
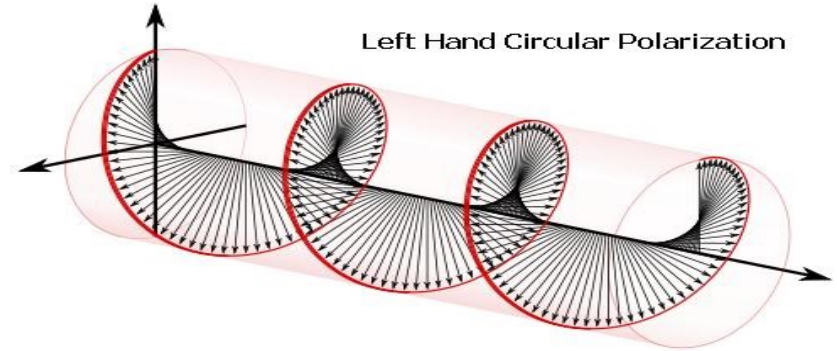
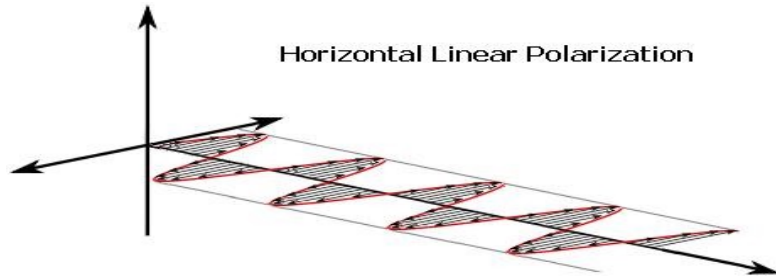
• Emitter



3D Polarization Technique (linear)



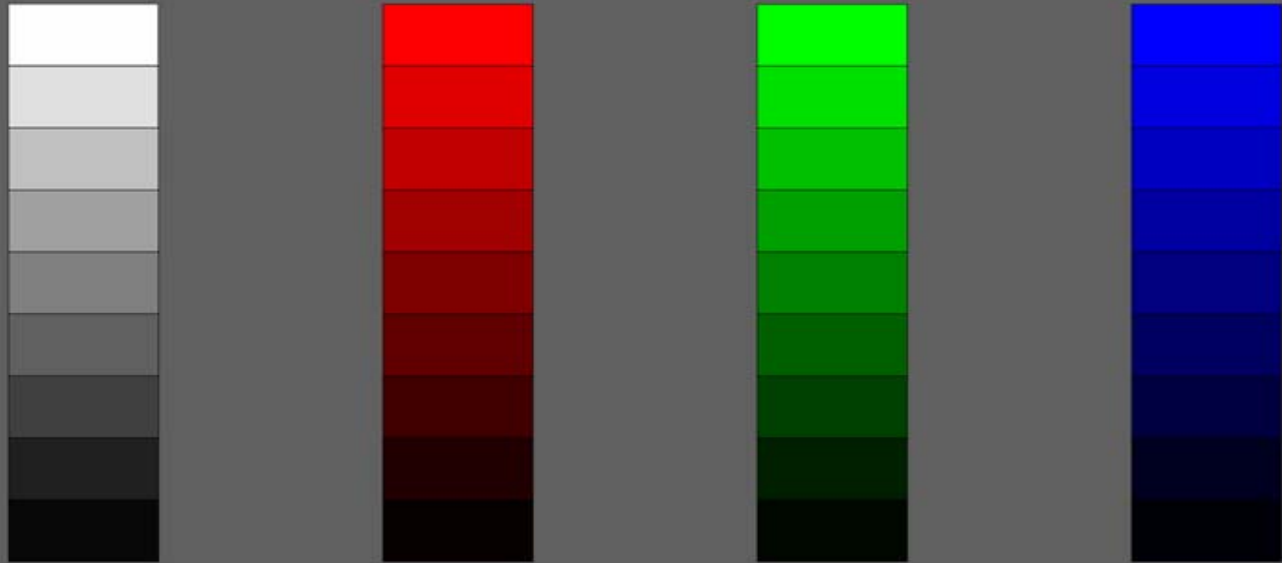
3D Polarization Technique (circular)



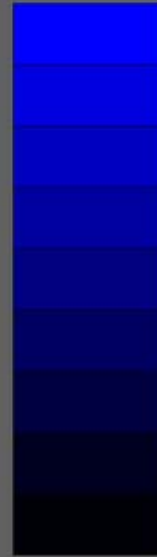
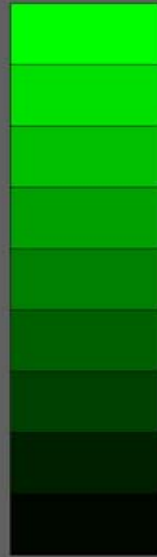
RealD Polarization



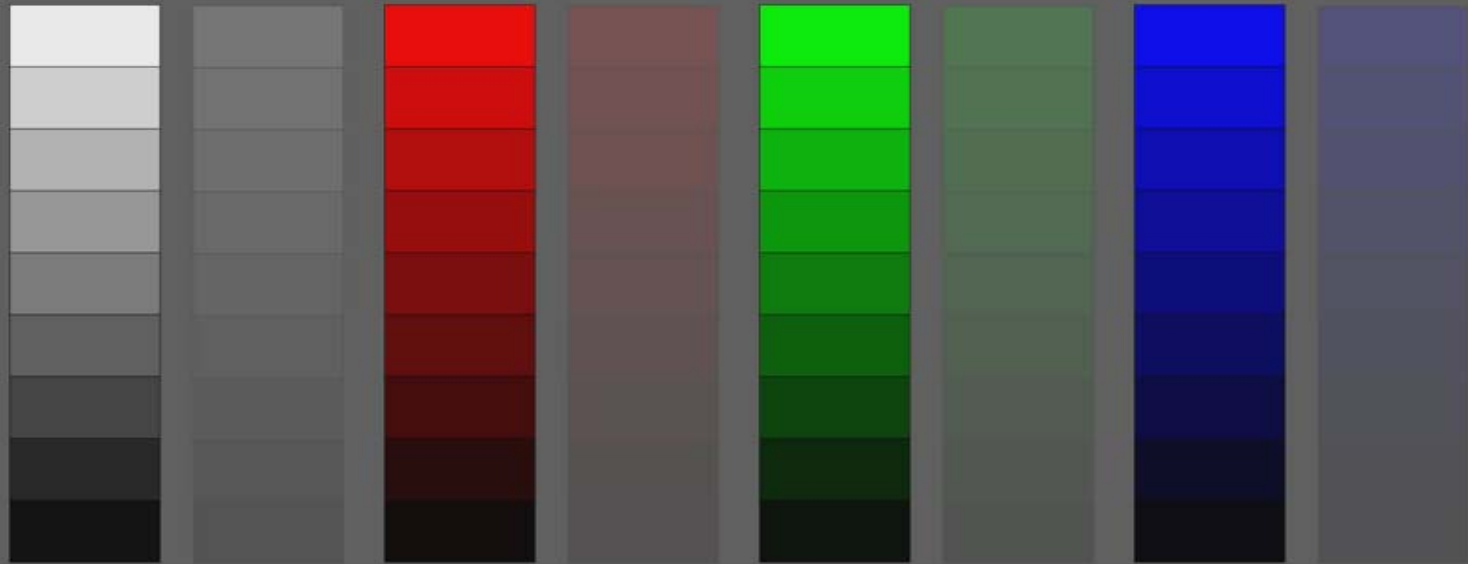
Ghosting: Original Left Image



Ghosting: Original Right Image



Ghost Image in Right Eye



Motion (24 fps)

- Simultaneous Motion
 - Left and Right eyes want motion to appear at same time
 - Sequential frame 3D naturally offsets motion timing
 - Parallax changes frame to frame
 - Double or triple flash required to improve motion
 - More flashes = better simultaneity = better parallax uniformity



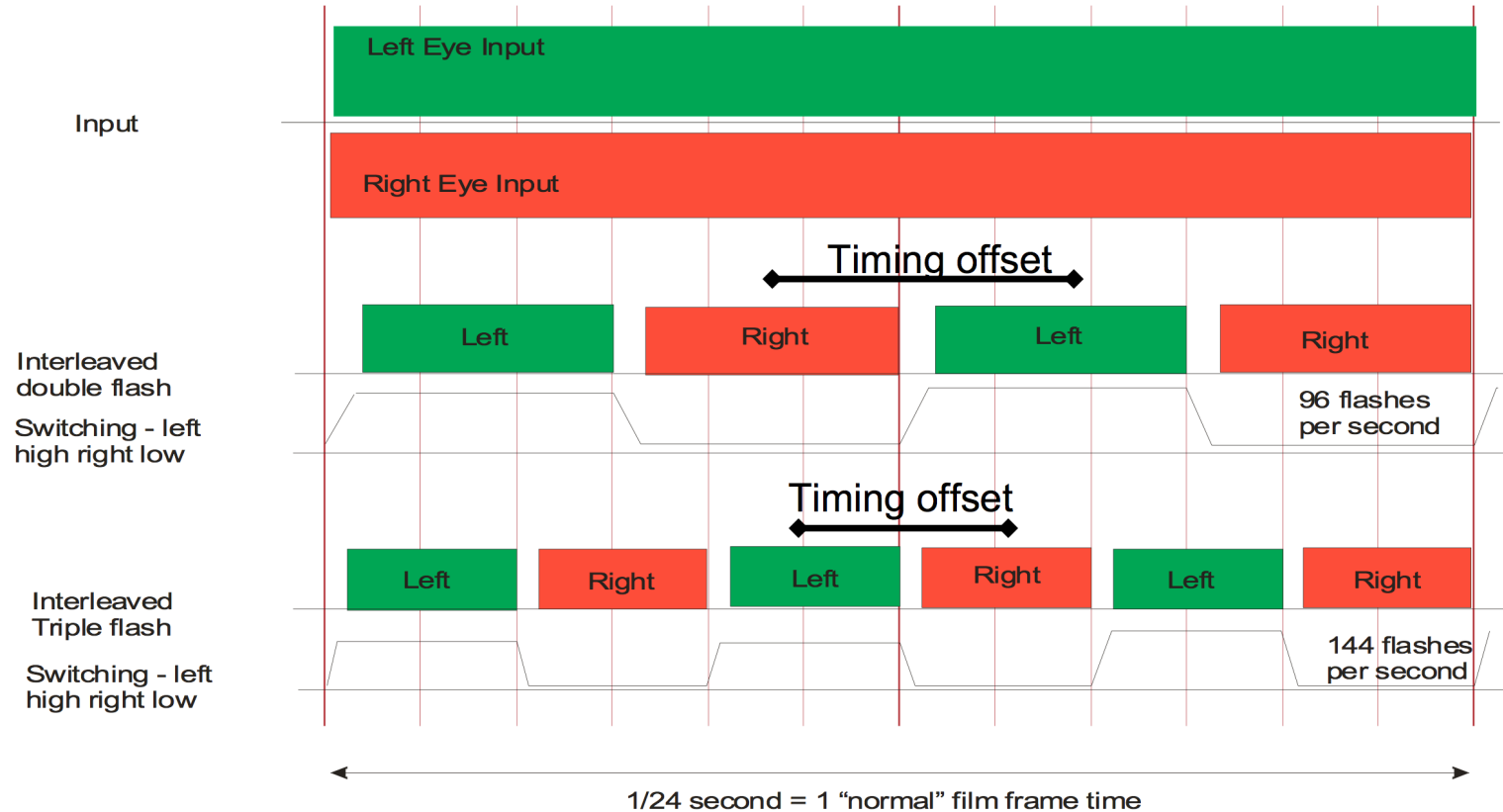
R



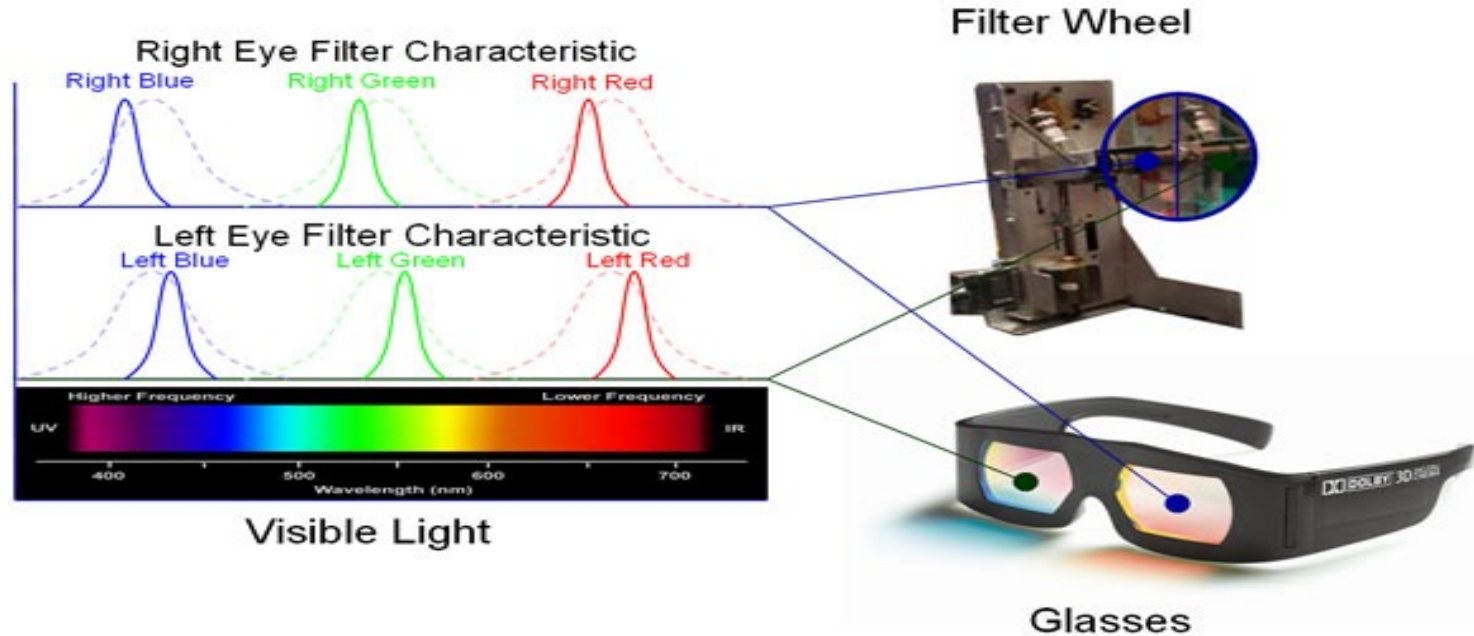
Motion – Double Flash, Tripple Flash

- Single projector systems
 - “Double Flash” – 96fps presentation
 - L1, R1, L1, R1, L2, R2, L2, R2 etc
 - “Triple Flash” – 144 fps presentation
 - L1, R1, L1, R1, L1, R1, L2, R2, L2, R2, L2, R2 etc.
 - Triple flash exceeds the fusion threshold for much smoother motion
 - Triple flash provides less phase difference between left and right eye views.

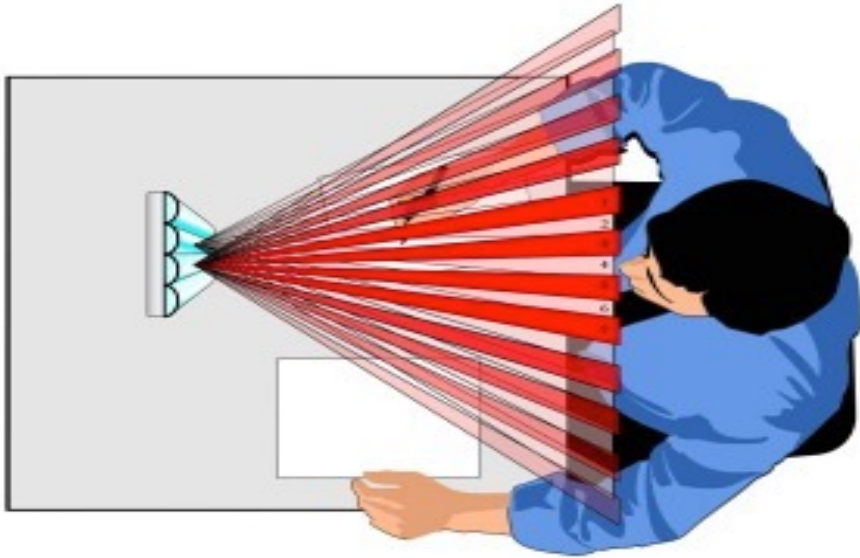
Timing Diagramm for Double and Tripple Flash Interleaved 3D



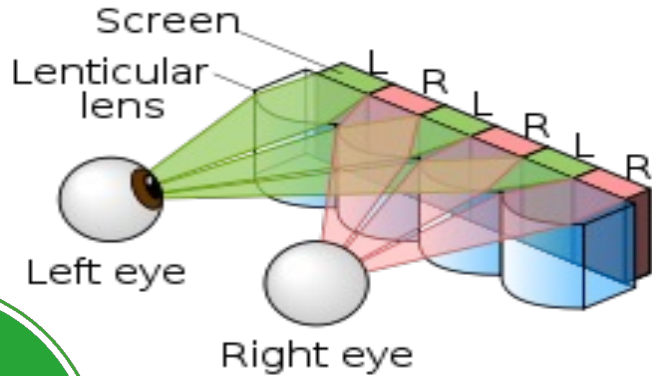
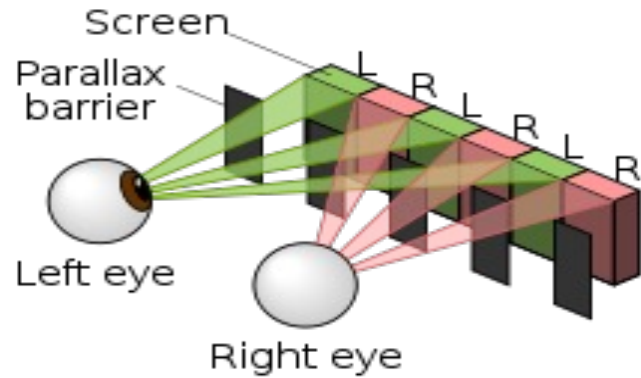
Infitec Technology (→ Dolby 3D)



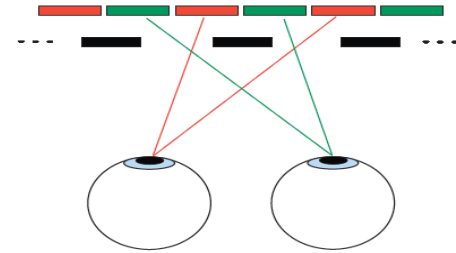
Naked (Glasses-free) 3D



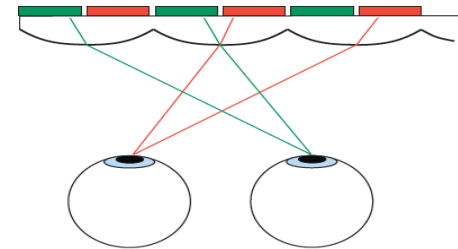
Naked (Glasses-free) 3D



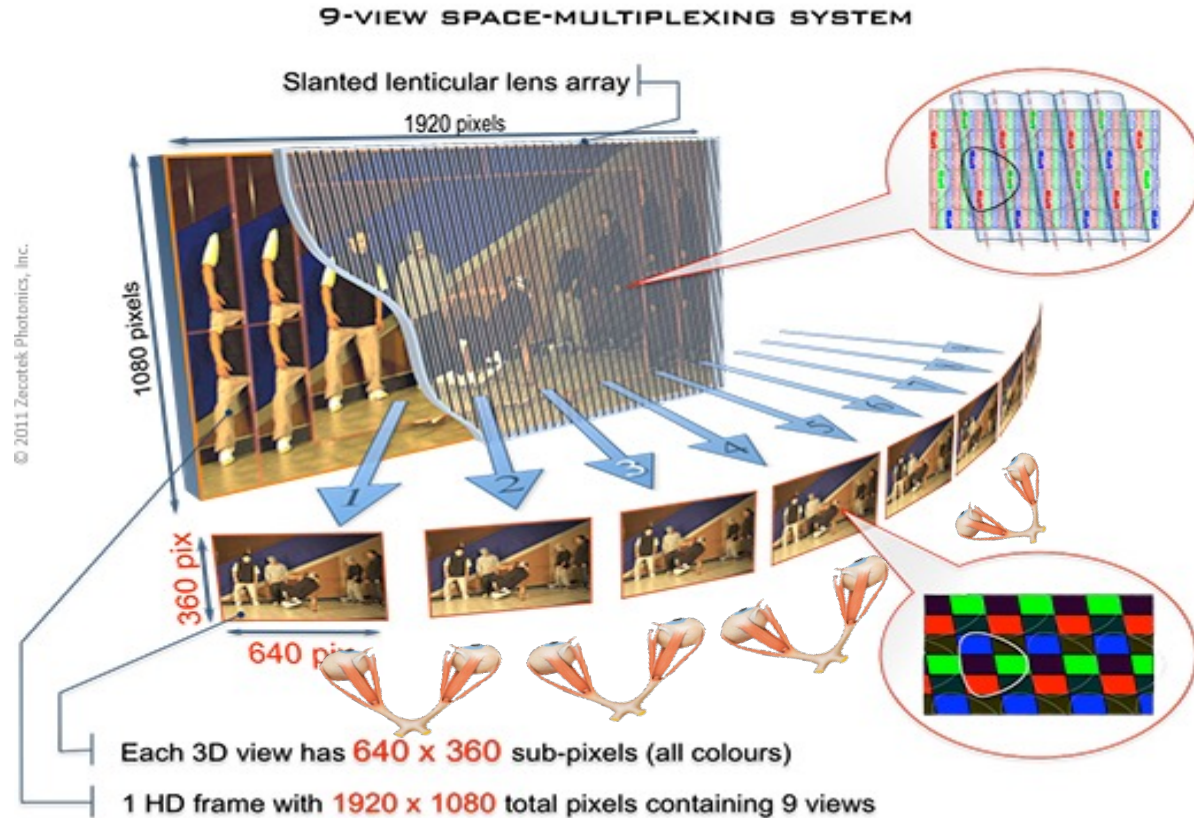
Parallax-Barrier Display



Lenticular Display



Slanted Lenticular Lens Array



3D Displays Comparison

Property	Stereo Display	Multi-View Display
Viewing Aid	Mostly required	Not required
“Look around” Effect	No	Yes
Resolution per View	High	Low
Perceived Scene Depth	High	Low

Lernziele

- Was ist Stereoskopie?
- Wie kann man Stereo-3D Inhalte darstellen?



Vielen Dank!

Sebastian Knorr

